

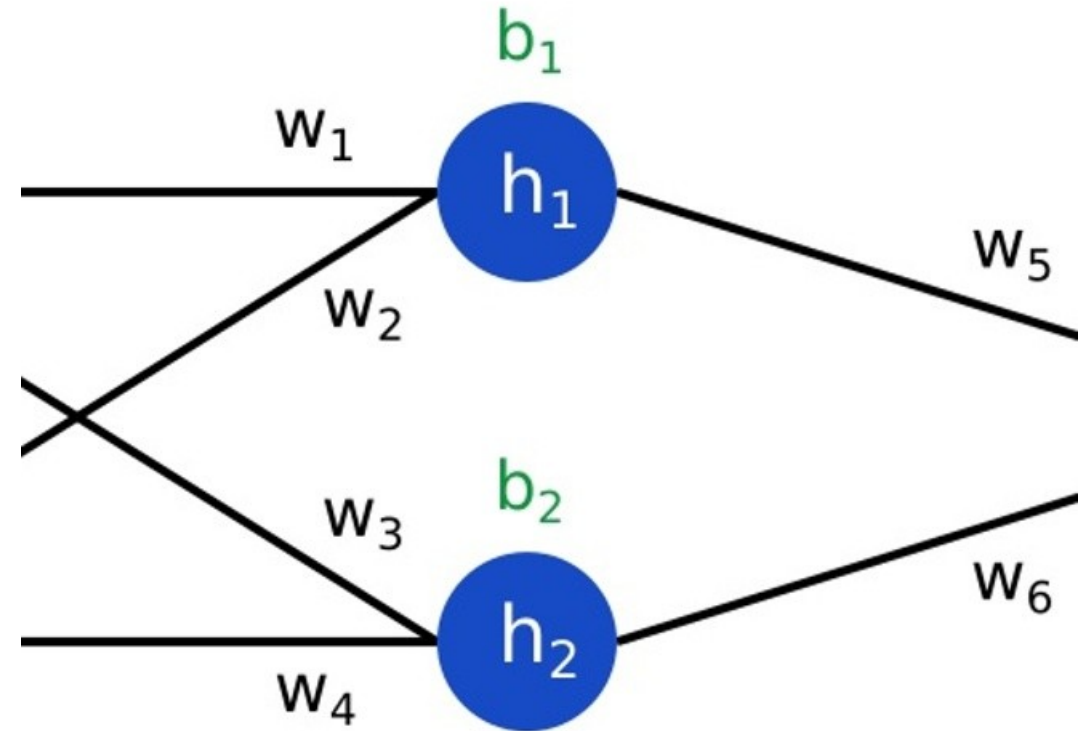
# Neural Networks: The Engine of M

## Hidden Layer

Understanding the basics

### KEY TAKEAWAY

Neural networks power many AI breakthroughs.



# What is a Neural Network?

Layers of interconnected neurons

**Neural networks learn by adjusting weighted connections across layers.**

# Key Performance Milestones

From early perceptrons to today

**15.3%**

**ImageNet top-5 error (2012)**

AlexNet reduced error to 15.3% in 2012

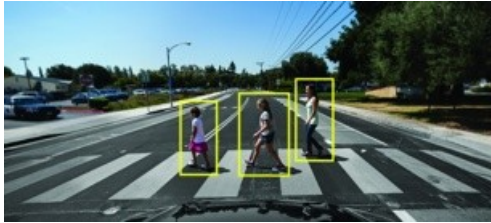
**3.6%**

**ImageNet top-5 error (2015)**

ResNet-152 achieved 3.6% top-5 error in 2015

# Real-World Applications

Where neural networks are used



## Computer Vision

Object detection, image classification, facial recognition



## Natural Language Processing

Chatbots, translation, sentiment analysis



## Speech Recognition

Transcribing audio, virtual assistants



## Healthcare

Medical imaging diagnosis, drug discovery

# Takeaways

Core concepts to remember

**Neural networks learn patterns through layered computation.**